

Demonstration of 169-Gbaud Faster-Than-Nyquist PAM4 Transmission Using Adaptation-Free THP-FFDNN Under 37.9-GHz 3-dB Bandwidth Limitation

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Abstract We propose a THP-FFDNN framework for faster-than-Nyquist PAM4 transmission, which eliminates adaptive updating while requiring only 408 parameters. The proposed scheme is compatible with GPU-based DSP implementation and achieves 169 Gbaud PAM4 under 37.9-GHz 3-dB bandwidth limitation, supporting the lowest spectral compression ratio of 62.5%. ©2026 The Author(s)

Introduction

4-level pulse-amplitude modulation (PAM4) has been widely adopted in short-reach optical interconnects due to its low complexity and compatibility with existing interfaces [1]. However, as the bandwidth scaling of electronic and optical components starts to lag behind the rapidly increasing capacity demand, whether PAM4 can still support higher-baud-rate optical interfaces under limited bandwidth remains questionable [2]. One possible solution is to employ higher-order modulation to improve spectral efficiency, but this often sacrifices compatibility with existing device and protocol interfaces [3]. Alternatively, faster-than-Nyquist (FTN) signaling provides a promising route to exceed the Nyquist bandwidth limit and enable higher-baud-rate PAM4 transmission within a constrained bandwidth [4].

Most existing FTN PAM4 schemes, however, still rely on cumbersome digital signal processing (DSP) pipelines including clock recovery and adaptive equalization, requiring additional DSP functions or dedicated modules [2-4]. This is less desirable in emerging artificial intelligence-oriented computing infrastructures, where signal processing is increasingly expected to be executed directly on edge graphics processing units (GPUs) instead of conventional DSP modules [5], as illustrated in Fig. 1. Such a trend makes neural-network-based transceiver design particularly attractive [6, 7].

In this paper, we propose a Tomlinson-Harashima-precoder-cascaded feed-forward decoder neural network (THP-FFDNN) framework for FTN PAM4 transmission. Unlike conventional schemes relying on clock recovery and adaptive

DSP, the proposed framework enables FTN transmission with fixed deployed parameters, thereby removing the need for pilot-assisted adaptation before operation. The proposed scheme requires only 408 parameters, making it highly lightweight and well suited to GPU-based signal recovery. Experimentally, we achieved 169-Gbaud FTN PAM4 transmission over a 0.5-km C-band single-mode fiber (SMF) link over a 37.9-GHz 3-dB bandwidth-limited channel and the method proposed supports the lowest spectral compression ratio of 62.5%.

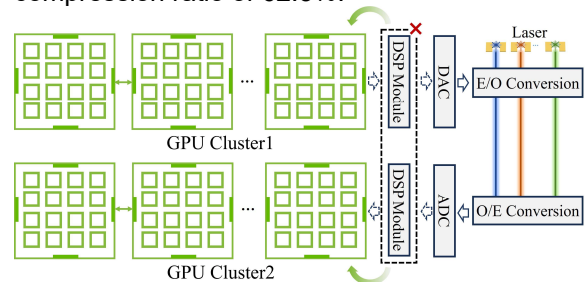


Fig. 1: Interconnection of computing centers without relying on traditional DSP modules.

Principle

Figure 2 illustrates the proposed THP-FFDNN for FTN PAM4 transmission. The framework is first jointly trained as a decision-feedback decoder neural network (DFDNN) to mitigate the ISI introduced by FTN filtering and the bandwidth-limited channel, and is then deployed as a transmitter-side Tomlinson-Harashima precoder neural network (THPNN) together with a receiver-side feed-forward decoder neural network (FFDNN). The design is inspired by the Tomlinson-Harashima

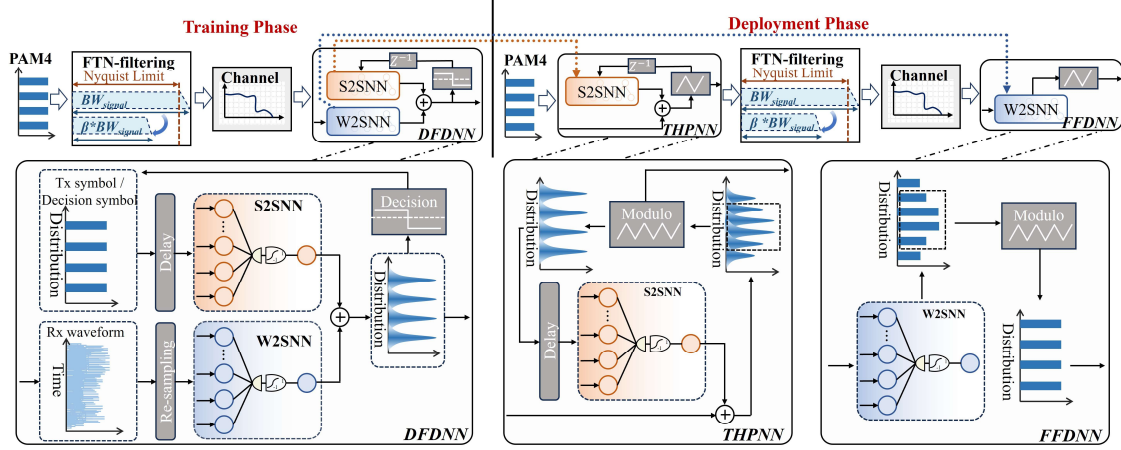


Fig. 2: Schematic of the proposed THP-FFDNN framework for FTN PAM4 transmission.

precoding principle that causally known interference can be compensated at the transmitter rather than cancelled at the receiver through decision feedback. For FTN PAM4 transmission, the major post-cursor ISI is determined by previously transmitted symbols and is therefore predictable at the transmitter. This allows the learned symbol-domain interference to be pre-compensated before transmission, thus avoiding the error propagation inherent to conventional receiver-side decision-feedback equalization.

In the training phase, the DFDNN consists of a symbol-to-symbol neural network (S2SNN) and a waveform-to-symbol neural network (W2SNN). The S2SNN takes delayed transmitted symbols during training, or decided symbols during inference, and learns the symbol-domain interference induced by FTN filtering and channel distortion. In parallel, the W2SNN operates on the received waveform and learns the waveform-to-symbol mapping for PAM4 recovery. Their outputs are jointly combined to reconstruct the target symbol, enabling offline end-to-end optimization.

After training, the learned S2SNN is transferred to the transmitter and embedded into a THP-like neural precoding loop as THPNN. Using previously precoded symbols as feedback, the THPNN predicts and pre-cancels the causal ISI prior to FTN filtering and transmission, while the modulo operation constrains the precoder output. Meanwhile, the trained W2SNN is kept at the receiver as the FFDNN for waveform-to-symbol detection and residual distortion compensation. Since both THPNN and FFDNN directly inherit the weights of the offline-trained DFDNN, all parameters remain fixed during deployment. Therefore, unlike conventional FFE/DFE-based receivers, the proposed THP-FFDNN requires no pilot-assisted adaptive updating before each inference. Moreover, all subnetworks are implemented using simple fully connected layers, with only 408 trainable parameters in total.

Experimental Setup

Figure 3 depicts the experimental setup used to evaluate FTN PAM4 transmission with different methods, including traditional DSP based on FFE or DFE, DFDNN, and THP-FFDNN. The input bit sequence is first processed by the DSP corresponding to the method under test to generate transmitting waveform. The digital signal is converted into an analog signal by an arbitrary waveform generator (AWG, Keysight M8199B, 80 GHz). Then the generated signal is amplified by an electrical amplifier (EA). The resulting analog waveform drives a thin-film lithium niobate Mach-Zehnder modulator (TFLN-MZM, 110 GHz), which is fed by a 1550-nm continuous-wave laser. The modulated optical signal is transmitted either back-to-back (B2B) or over 0.5 km of SMF. At the receiver, a variable optical attenuator (VOA) is utilized prior to the photodetector (PD, 100-GHz) to fine-tune the received optical power. The detected electrical signal is sampled to complete analog-to-digital conversion via oscilloscope (OSC, Keysight UXR- 0594BP, 59-GHz) to obtain receiving waveform.

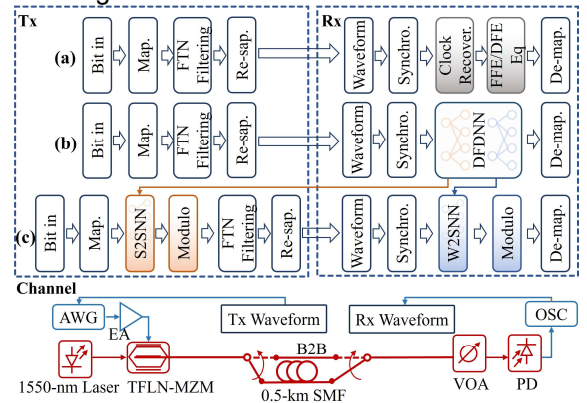


Fig. 3: The DSP flow of (a) traditional method, (b) DFDNN and (c) THP-FFDNN with the specific experimental setup for FTN PAM4 testing system.

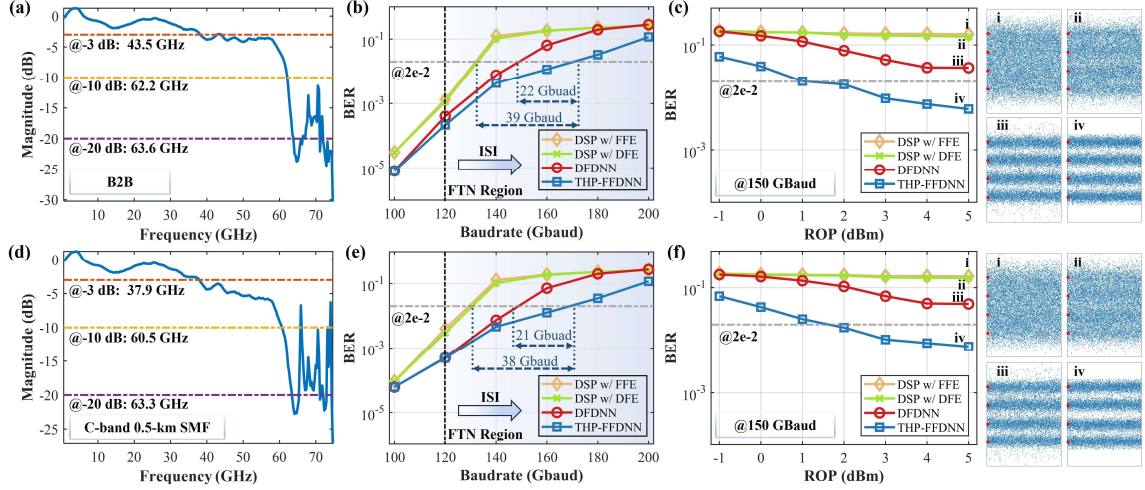


Fig. 4: The system response and BER performance of different methods versus baudrate and ROP under B2B conditions (a, b, c) and 0.5-km SMF conditions (d, e, f). Insets: recovered PAM4 symbols before decision of different methods.

Experimental Results and Discussions

Figure 4 presents the detailed system responses, and BER performance of different methods versus baud rate and received optical power (ROP) under B2B and 0.5-km SMF conditions, respectively. As shown in Fig. 4(a), the measured back-to-back bandwidths of the system are 43.5 GHz, 62.2 GHz, and 63.6 GHz at the -3 dB, -10 dB, and -20 dB points, respectively, which is mainly limited by the oscilloscope, with rapid attenuation observed beyond approximately 60 GHz. To achieve FTN transmission, the FTN signal bandwidth was fixed at 60 GHz throughout the experiments. In particular, the roll-off factor was set to $\alpha = 0.2$, and the spectral compression ratio β was determined by

$$(1 + \alpha) \cdot \beta \cdot \frac{\text{baud rate}}{2} = 60 \text{ GHz}. \quad (1)$$

For example, for baud rates of 100, 120, 140, 160, 180, 200 Gbaud, β was set to 100/100, 100/120, 100/140, 100/160, 100/180, 100/200.

Fig. 4(b) shows the BER performance of different methods versus baud rate. As the baud rate increases, FTN PAM4 signals suffer from more severe ISI under bandwidth-limited conditions, causing rapid BER degradation for conventional DSP schemes based on FFE or DFE. In contrast, the proposed THP-FFDNN effectively mitigates ISI and achieves baud-rate gains of 39 Gbaud and 22 Gbaud over DSP with DFE and DFDNN, respectively, under the 20% SD-FEC threshold of 2×10^{-2} . As shown in Fig. 4(c), THP-FFDNN further achieves a receiver sensitivity of 1.01 dBm at 150 Gbaud, while the other methods fail to reach the FEC threshold.

Fig. 4(d) shows the system response after 0.5-km SMF transmission. The measured bandwidths are 37.9 GHz, 60.5 GHz, and 63.3 GHz at the -3 dB, -10 dB, and -20 dB points,

respectively, indicating slight degradation due to dispersion. As shown in Fig. 4(e), THP-FFDNN still maintains a clear advantage, supporting up to 169 Gbaud and achieving baud-rate gains of 38 Gbaud and 21 Gbaud over DSP with DFE and DFDNN. The results also show the lowest spectral compression ratio of 62.5% (100/160) can be achieved under the FEC threshold of 2×10^{-2} . Fig. 4(f) further shows, for the 150-Gbaud signal, THP-FFDNN achieves a receiver sensitivity of 1.57 dBm, whereas the BER of other methods remains above the FEC threshold.

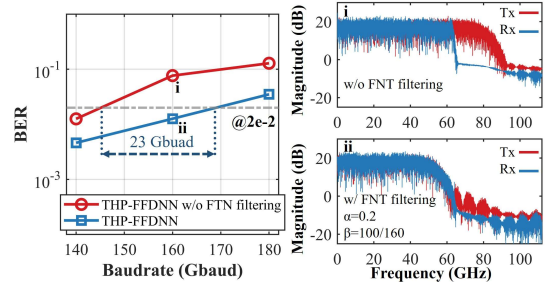


Fig. 5: BER performance of THP-FFDNN without or with FTN filtering on the transmission end. Insets: the spectrum of signals without or with FTN filtering at 160 Gbaud.

We further evaluate the effect of FTN filtering on performance. Fig. 5 shows the BER results of THP-FFDNN without and with FTN filtering. As observed, by applying FTN filtering, uncontrolled signal truncation induced by the limited channel bandwidth can be avoided, leading to improved signal recovery at the receiver.

Conclusions

A THP-FFDNN framework is proposed for faster-than-Nyquist PAM4 transmission, which eliminates adaptive updating while requiring only 408 parameters. The proposed scheme is compatible with GPU-based DSP implementation and experimentally achieves 169 Gbaud PAM4 over a 37.9-GHz 3-dB bandwidth-limited channel.

Acknowledgements

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